# AI & MACHINE LEARNING

&

# INTERNET OF THINGS

Combined Section-wise Study Notes

**Part A — AI & ML:** artificial intelligence · types · evolution · agents · domains · search · machine learning · blockchain · fuzzy logic · generative AI

**Part B — IoT:** architecture · requirements · Web 3.0 & WoT · middleware & protocols · standardization · WSN · SCADA & RFID · cloud, grid, SoA, MCC · IIoT · ARMET

Typed from the two sets of handwritten class notes — every topic, example and diagram retained, with clarifications where a page was unclear or needs a correction.

# Contents


**A1. Artificial Intelligence** 5

A1.1 AI programming languages . . . 5

A1.2 Branches of AI . . . 5

**A2. Types of AI** 5

A2.1 Based on capability . . . 6

A2.2 Based on functionality . . . 6

**A3. Evolution of AI** 6

**A4. Agents** 8

A4.1 Five types of agents . . . 8

**A5. Domains of AI and Searching Algorithms** 9

A5.1 Domains of AI . . . 9

A5.2 Searching algorithms . . . 9

**A6. Machine Learning** 10

**A7. Blockchain** 10

A7.1 Content of a block . . . 11

A7.2 Features . . . 11

A7.3 History of blockchain . . . 11

**A8. Fuzzy Logic** 11

**A9. Generative AI** 12

**B1. Internet of Things — Basics** 14

B1.1 Functional requirements . . . 14

B1.2 Architecture of IoT (4 layers) . . . 14

B1.3 Four pillars and the DNA of IoT . . . 14

B1.4 Views of IoT . . . 15

B1.5 Ubiquity of IoT (present everywhere) . . . 15

**B2. Web 3.0 and the Web of Things** 15

B2.1 Web 3.0 . . . 15

B2.2 WoT — Web of Things . . . 15

**B3. Middleware, Protocols and Standardization** 17

B3.1 Middleware in IoT . . . 17

B3.2 Protocols used . . . 17

B3.3 Protocol standardization . . . 17

B3.4 Machine-to-machine (M2M) protocols . . . 17

**B4. WSN, IEEE 802.15.4, SCADA and RFID** 18

B4.1 WSN protocols . . . 18

B4.2 IEEE 802.15.4 . . . 18
B4.3 SCADA and RFID . . . 18
**B5. Cloud, Grid, SoA, MCC, IIoT and ARMET** **18**
B5.1 CoT — Cloud of Things . . . 18
B5.2 Grid computing . . . 18
B5.3 SoA — Service Oriented Architecture . . . 18
B5.4 Cloud computing and cloud standards . . . 19
B5.5 MCC — Mobile Cloud Computing . . . 19
B5.6 IIoT — Industrial Internet of Things . . . 19
B5.7 Basic technologies used in IoT . . . 19
B5.8 ARMET . . . 19
B5.9 Role of IoT in a collaborative production environment . . . 20
**Appendix — Clarifications and reading notes** **21**

PART A

# Artificial Intelligence & Machine Learning

AI · Agents · Search · ML · Blockchain · Fuzzy logic · Generative AI

# A1. Artificial Intelligence

- **AI** is the branch of computer science in which machines are developed that can **perform a wide range of tasks expertly** and do not require human intelligence at work (as written — i.e. they do the task without a human doing it).
- In **1956**, **John McCarthy** first adopted the term AI — therefore he is the **father of AI**.
- **Humanoid** = human-like but not human; developed to make these humanoids work.
- In **1958** John McCarthy developed the first AI language, **LISP** (List Programming).
- The AI term includes: **learning, reasoning, problem solving**.

## A1.1 AI programming languages

| Language | Note |
|---|---|
| **LISP** | List Programming — the **first AI language** (1958, John McCarthy) |
| **Prolog** | Programming in Logic |
| Others | Java, Python, R language, Julia, Smalltalk, Haskell, C++, Scala etc. |

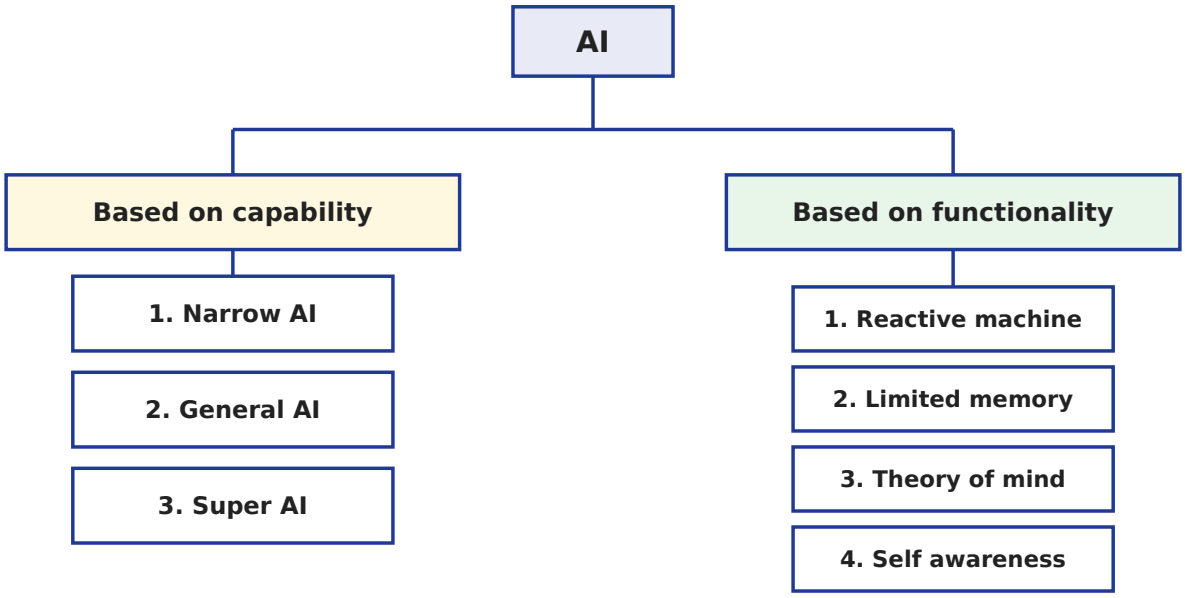
## A1.2 Branches of AI

| | Branch |
|---|---|
| (a) | Machine learning (including deep learning) |
| (b) | Expert system |
| (c) | Fuzzy logic |
| (d) | Neural networks (written “neutral logic”) |
| (e) | Natural language (processing) |
| (f) | Robotics |

**Remember**

**First AI program = The Logic Theorist**, by **Allen Newell** (the year is hard to read in the notes — about 1956/57).

# A2. Types of AI




*Types of AI: based on capability and based on functionality (tree from the notes).*

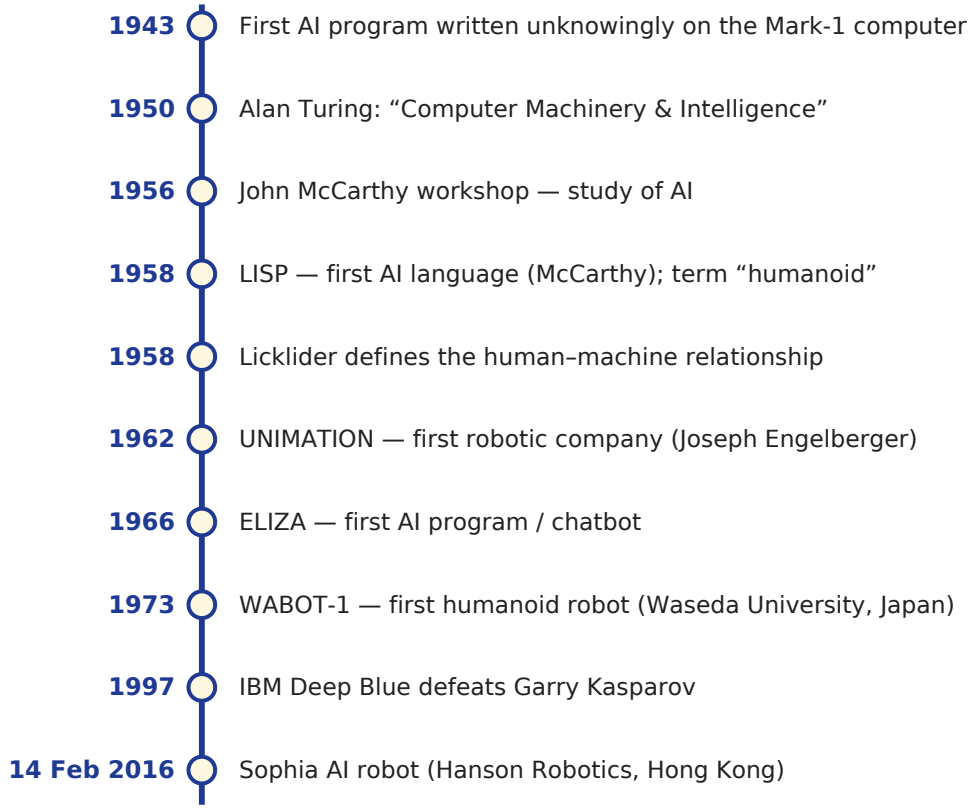
### A2.1 Based on capability

| Type | Meaning | Examples / status |
|---|---|---|
| **(1) Narrow AI** | Does a **specific task** | (a) Virtual / digital assistants — Alexa, Siri, Cortana, Bixby, Google Assistant<br>(b) Face recognition / image recognition systems<br>(c) Tesla self-driving car<br>(d) Email spam filter |
| **(2) General / Strong AI** | Can do work **exactly like a human being** | Still in the **development phase** |
| **(3) Super AI** | **Stronger than human beings** | Still in the **imaginary phase** |

### A2.2 Based on functionality

| Type | Meaning | Example |
|---|---|---|
| **(1) Reactive machine** | Focuses on the **present scenario**; ignores past experience and memory | IBM chess player **Deep Blue (1997)** — defeated the rank-1 chess player Garry Kasparov |
| **(2) Limited memory** | **Does not ignore** past experience and memory | Self-driving car |
| **(3) Theory of mind** | AI will **work as human beings** | — |
| **(4) Self awareness** | **Machines stronger than human beings** | — |

## A3. Evolution of AI




*Timeline of AI as written in the notes.*

| Year | Event |
|---|---|
| 1943 | First AI program written **unknowingly** on the **Mark-1** computer |
| 1950 | Alan Turing launched the book / article **“Computer Machinery & Intelligence”** |
| 1956 | **John McCarthy** workshop — study of AI |
| 1958 | **LISP** (List Programming) — first AI language, developed by John McCarthy; introduced the term “Humanoid” |
| 1958 | Licklider defines the **human–machine relationship** for the first time |
| 1962 | Started the first robotic company **UNIMATION** by **Joseph Engelberger** — father of robotics |
| 1966 | **ELIZA** — first AI program (chatbot) |
| 1973 | **WABOT-1** — first humanoid robot, Waseda University, Japan |
| 1997 | **IBM Deep Blue** — chess player; defeated the world’s rank-1 player Garry Kasparov |
| 14 Feb 2016 | **Sophia** AI robot invented — face recognition; Hong Kong company **Hanson Robotics**; first robot to get citizenship (2017) |

**Clarification**

Corrections to check: ELIZA was written by **Joseph Weizenbaum** (the notes name Joseph Engelberger, who founded Unimation); Sophia received **Saudi Arabian** citizenship in 2017 (the notes say UAE); Deep Blue was a chess computer, not a humanoid robot (the notes call it the “first intelligent humanoid”).

# A4. Agents

- An **agent** is an independent entity that **observes the environment to achieve a goal**.

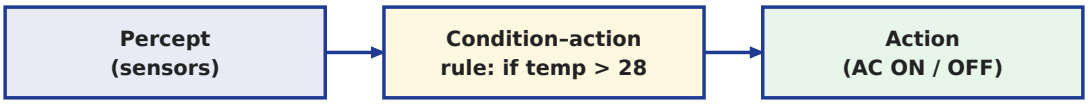
**Formula**

**Agent = Architecture + Program**

## A4.1 Five types of agents

### (a) Simple reflex agent

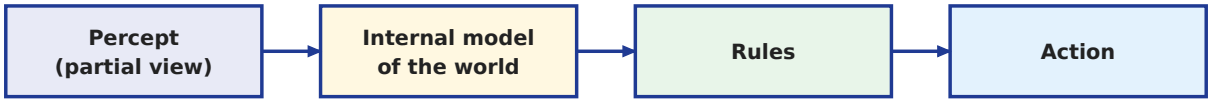
- Observes the environment and **takes action according to a condition**; **pre-programmed**.
- Example: **if temp > 28 → AC ON, else AC OFF**.




*Simple reflex agent (flow added from the description).*

### (b) Model-based reflex agent

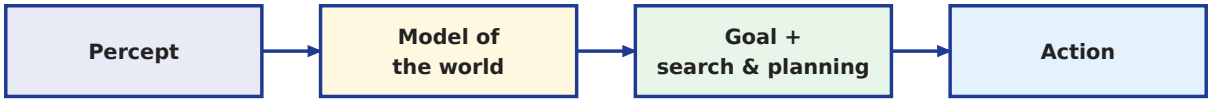
- Observes the environment **partially** and takes action.
- Example: self-driving cars.




*Model-based reflex agent.*

### (c) Goal-based agent

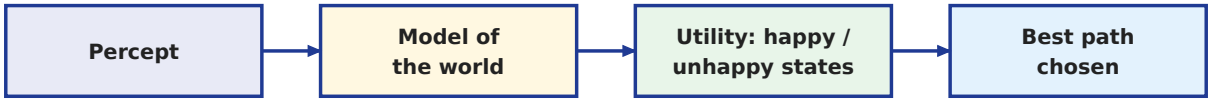
- The **extended version of the model-based reflex agent**; uses **searching and planning**.
- Example: **Google Waymo** self-driving car (2009).




*Goal-based agent.*

### (d) Utility-based agent

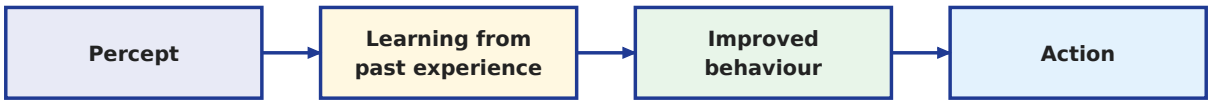
- A **happy / unhappy state** based agent; chooses the **optimized path from multiple paths**.
- Example: price comparison.




*Utility-based agent.*

### (e) Learning agent

- **Learns from past experience** — like human beings.




*Learning agent.*

| Agent | Decides by | Example |
|---|---|---|
| Simple reflex | A fixed condition → action rule | AC on when temp > 28 |
| Model-based reflex | Condition + partial view of the world | Self-driving car |
| Goal-based | Searching and planning towards a goal | Google Waymo (2009) |
| Utility-based | Happy / unhappy states — best path | Price comparison |
| Learning | Past experience | Human beings |

*Summary table and the agent flow pictures are added for revision; the notes give the definitions and examples above.*

# A5. Domains of AI and Searching Algorithms

## A5.1 Domains of AI

| Domain | Meaning | Example |
|---|---|---|
| **(A) Data science** | Study of data — can be **audio, video or textual**; data systems & processing; **mathematical and statistical principles** are applied | — |
| **(B) Computer vision** | **Images and objects are identified** | Face-recognition apps (Google Lens, Google Translate) |
| **(C) NLP** — Natural Language Processing | Used for **communication between computers and human beings** | — |

## A5.2 Searching algorithms

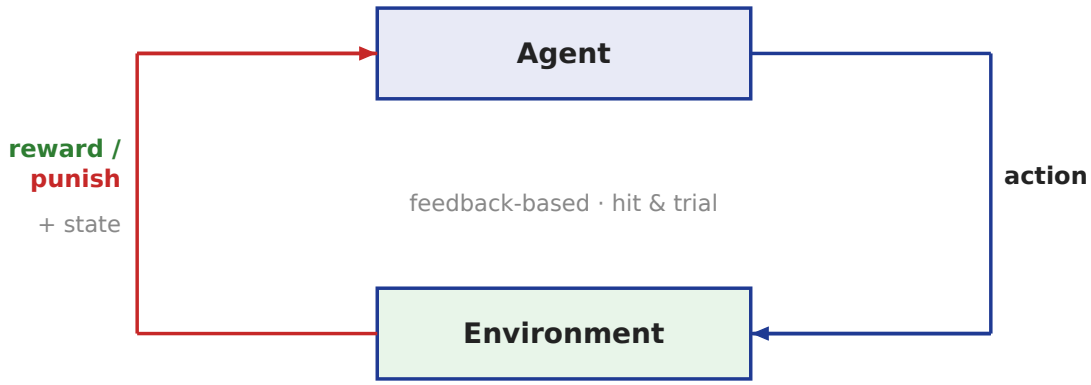
| Informed search | Uninformed search |
|---|---|
| Heuristic searching algorithm<br>Knowledge-based<br>Solution is found **quickly**<br>No guarantee of completion<br>Direction to the solution is **known**<br>Examples: **A*, graph search** | Blind / brute-force searching algorithm<br>Does **not use knowledge**<br>Solution is found **late**<br>Guarantee of completion<br>Direction is **unknown**<br>Examples: **BFS, uniform-cost search, DFS** |

# A6. Machine Learning

- ML is a **subsystem (subset) of AI**.
- ML **learns by itself**, so there is no need to write a separate algorithm / program.
- **3 types:** (A) supervised, (B) unsupervised, (C) reinforcement learning.

| | (A) Supervised | (B) Unsupervised | (C) Reinforcement |
|---|---|---|---|
| Data | Input data given **along with labels** (details about the data) | **Unlabelled** data | No fixed data — the agent acts in an environment |
| How it learns | Learns from the labels | Data is **trained by itself**; **hidden patterns** are also found | **Trained by itself** — reward and punishment; feedback-based; hit & trial |
| Kinds / algorithms | (a) **Regression** — relationship between 2 or more variables; (b) **Classification** — works on Yes / No | **K-means** cluster algorithm, **hierarchical** cluster algorithm | **Value-based**, **policy-based** |

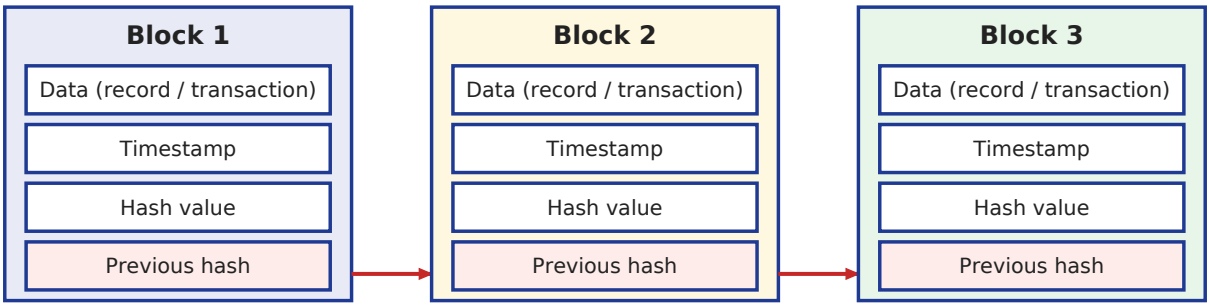
### Reinforcement learning — the agent / environment loop




*The agent acts on the environment and gets a reward or punishment (feedback) in return.*

# A7. Blockchain

- A **type of ledger** that keeps a record of the transactions happening in a **peer-to-peer network**.
- Any information transferred through blockchain is **encrypted / secured**, so it **cannot be changed**.
- In a block there is a **data structure for a digital transaction**; data / transactions are stored with a **hash value** so that no changes can be made to them.
- When multiple blocks are connected together, a **secured chain** is formed — called a **blockchain**.




each block stores the hash of the block before it — changing one block breaks the chain

*Blocks linked into a chain: every block holds the hash of the previous block (picture added from the description).*

### A7.1 Content of a block

| Field | Meaning |
|---|---|
| **(a) Data** | Record / transaction |
| **(b) Timestamp** | Creation time of the block |
| **(c) Hash value** | Block security |
| **(d) Previous hash** | Hash value of the previous block |

### A7.2 Features

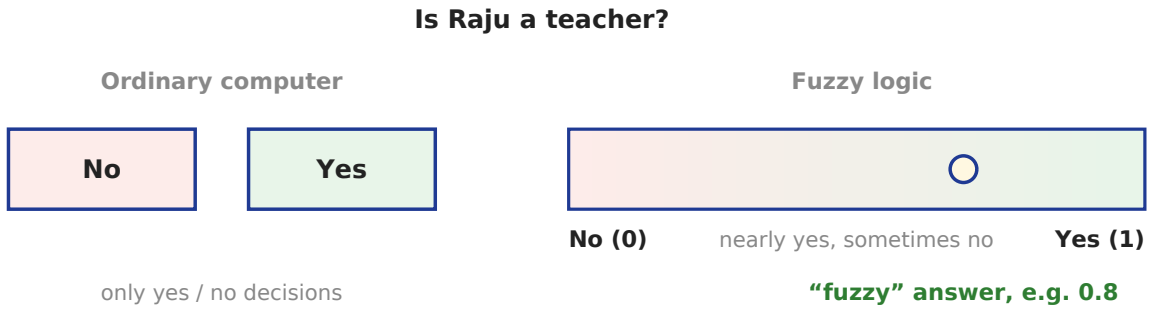
- **Data, records, contracts and documents** can be stored.
- There is **no privacy breach**.
- **No centralized control**.
- **No specific owner**.

### A7.3 History of blockchain

| Year | Event |
|---|---|
| 1991 | **Stuart Haber & Scott Stornetta** proposed the concept of a cryptographically secured block (secured blockchain) |
| 1992 | The first secured blockchain was created using a **Merkle tree** (Ralph Merkle, 1979) |
| 2004 | **RPOW** (Reusable Proof of Work) — a system created to store digital transactions; digital cash was stored; by **Hal Finney** |
| 2008 | **Satoshi Nakamoto** started cryptocurrency (Bitcoin) |

## A8. Fuzzy Logic

- A **method of reasoning similar to human reasoning**.
- **Fuzzy = not clear**; first introduced by **Lotfi Zadeh**.
- An ordinary computer only makes decisions as **yes / no**, but fuzzy logic also includes the **possibilities between yes and no**.




*Example from the notes: “Is Raju a teacher?” — yes, no, or a fuzzy answer in between (nearly yes, sometimes no).*

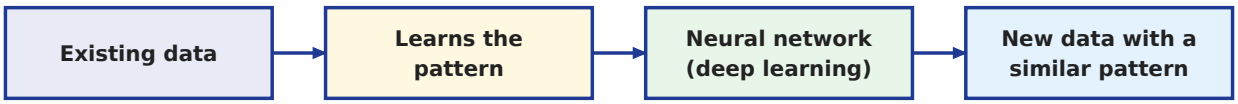
| Advantages | Disadvantages |
|---|---|
| Works with **any type of input**<br>Less memory needed<br>Structure is easy to understand<br>Provides efficient solutions to complex problems<br>Users can easily add or delete rules | The solution is **not always correct**<br>Sometimes very confusing<br>No systematic approach to its design<br>Very slow, so the output takes time |

### Applications

- Controlling the **height (altitude) of spacecraft and satellites**.
- **Decision-making support** systems.
- Assuming a **target in water**.
- **Cooking strategies in a microwave**.
- **Traffic control**.

## A9. Generative AI

- A technology that can **process existing data and create new data**.
- Based on **machine learning and deep learning**.
- It **learns the pattern in data** and generates new data with a similar pattern.
- Examples: **ChatGPT, DALL-E**.
- **Neural networks** are used, which work like the human mind.




*How generative AI works (flow added from the description).*

PART B

# Internet of Things

Architecture · Protocols · Standards · Cloud · IIoT

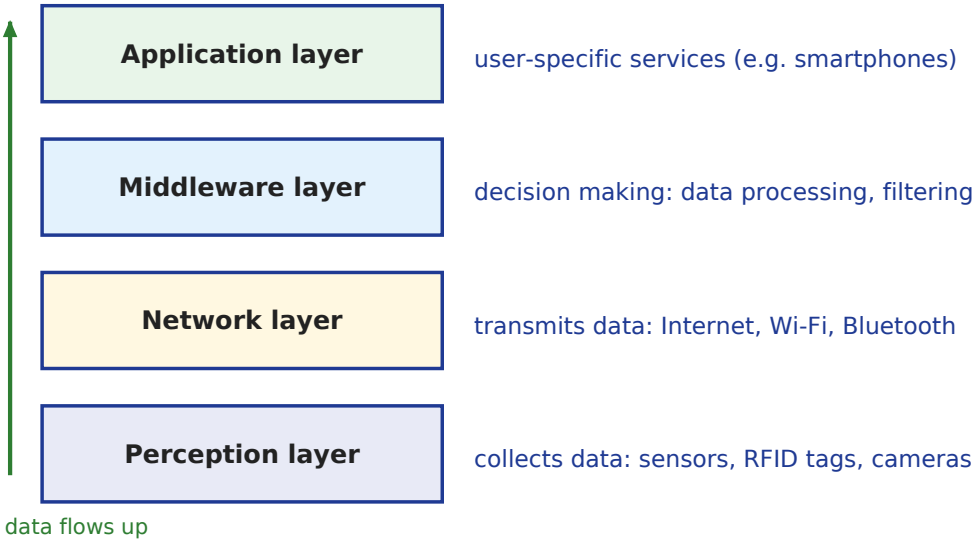
# B1. Internet of Things — Basics

- **IoT** is a type of network system in which **physical devices** such as sensors, gadgets, vehicles and home appliances are **connected to the internet**, allowing them to **exchange data**.

## B1.1 Functional requirements

| # | Requirement | Meaning |
|---|---|---|
| (i) | **Scalability** | The system should remain efficient even as the number of devices increases |
| (ii) | **Interoperability** | Devices from different manufacturers should be able to communicate with each other |
| (iii) | **Security** | Data should be secure and unauthorized access should not be allowed |
| (iv) | **Real-time monitoring** | The system should allow real-time monitoring and analysis of data |
| (v) | **Low-power consumption** | Devices should use minimal power |

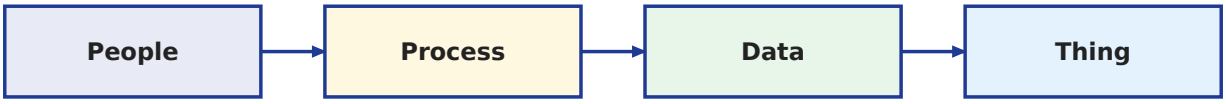
## B1.2 Architecture of IoT (4 layers)




*The four IoT layers (picture added from the list in the notes).*

| Layer | Job | Examples |
|---|---|---|
| **(i) Perception layer** | Collects data | Sensors, RFID tags, cameras |
| **(ii) Network layer** | Transmits data | Internet, Wi-Fi, Bluetooth |
| **(iii) Middleware layer** | Handles decision making | Data processing, filtering data |
| **(iv) Application layer** | Provides user-specific services | Smartphones |

## B1.3 Four pillars and the DNA of IoT




*The four pillars of IoT.*

| Four pillars | DNA of IoT |
|---|---|
| (i) People (ii) Process (iii) Data (iv) Thing | Data, Network, Application |

### B1.4 Views of IoT

| View | Focus |
|---|---|
| **(a) Technology-oriented** | Sensors, actuators, cloud computing, big data & AI |
| **(b) Application-oriented** | Smart phones, health care, agriculture |
| **(c) Service-oriented** | **IoTaaS** — IoT as a Service |

### B1.5 Ubiquity of IoT (present everywhere)

| Area | Use |
|---|---|
| **Smart homes** | Control automatically — light, AC, machines etc. |
| **Smart cities** | Traffic management, pollution control |
| **Health care** | Fitness bands, remote patient monitoring |
| **Agriculture** | Soil monitoring, smart irrigation |
| **Industrial IoT** (IIoT) | Machines, real-time monitoring |
| **Retail** | Inventory tracking, employee biometrics, smart checkout |

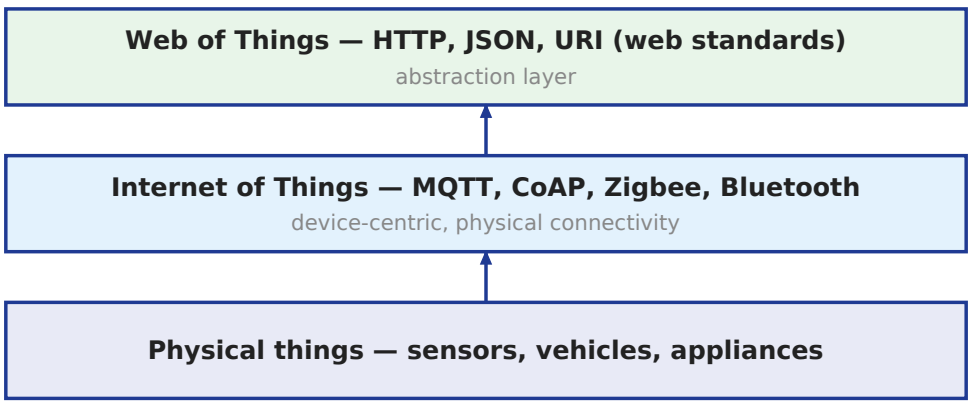
## B2. Web 3.0 and the Web of Things

### B2.1 Web 3.0

- A **semantic web** in which machines are developed to **understand and interpret data like human beings**.
- Web 3.0 + IoT enables **smart decision-making machines without human intervention**.
- Uses include **blockchain-enabled IoT security** and **DApps** (decentralised applications).

### B2.2 WoT — Web of Things

- An **extension of IoT**.
- Uses **web technologies (HTTP, JSON)** to allow devices to be easily accessed and integrated.
- The **abstraction layer built on top of IoT**, so that all devices can communicate using web standards and protocols.
- Example: if a **smart bulb has a URL**, it can be turned on / off directly from a web browser.




*WoT is an abstraction layer on top of IoT (picture added from the description).*

| Aspect | IoT | WoT |
|---|---|---|
| Connectivity | Device-centric and physical connectivity | Web-centric and application-layer abstraction |
| Protocol | MQTT, CoAP, Zigbee, Bluetooth | HTTP & JSON |
| Data formats | Binary & lightweight | JSON & XML supported |
| Interface | Mostly proprietary or protocol-specific | Standard web interface (URI) |
| Interoperability | Low — diverse protocols | High — via web standards |
| Goal | To connect the real world / objects | To embed IoT devices into the web |

# B3. Middleware, Protocols and Standardization

## B3.1 Middleware in IoT

- The **software layer that manages data between sensors and applications**.
- Functions: **(A) device management (B) data management (C) security handling (D) interoperability**.

## B3.2 Protocols used

| Protocol | Full form | Note |
|---|---|---|
| **(i) MQTT** | Message Queuing Telemetry Transport | **Lightweight**, low bandwidth |
| **(ii) CoAP** | Constrained Application Protocol | An **optimized web transfer protocol** |
| **(iii) HTTP / HTTPS** | Hypertext Transfer Protocol (Secure) | **Traditional internet communication** |
| **(iv) AMQP** | Advanced Message Queuing Protocol | — |

## B3.3 Protocol standardization

- To develop a **common set of rules** which all IoT devices can follow.
- Necessary so that devices from different manufacturers can communicate with each other.

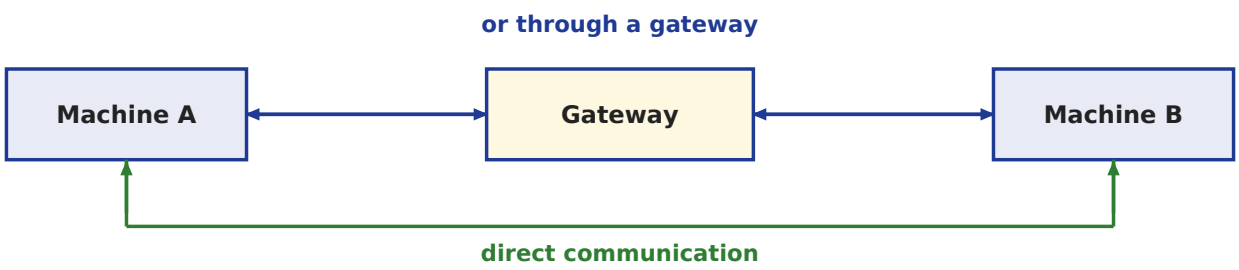
| Objectives |
|---|
| To ensure **interoperability** between different devices |
| To bring **scalability and flexibility** |
| To maintain **security and data integrity** |

### Efforts for standardization

| Body | Full name | What it does |
|---|---|---|
| **(i) IEEE** | Institute of Electrical & Electronics Engineers | Defines protocols for the **physical and data-link layers** |
| **(ii) IETF** | Internet Engineering Task Force | Develops **network-layer protocols**, such as **6LoWPAN** (low-power wireless personal area network) |
| **(iii) ETSI** | European Telecommunications Standards Institute | Standardizes **M2M** (machine-to-machine) protocols |

## B3.4 Machine-to-machine (M2M) protocols

- Machines communicate **directly or through a gateway**.
- Examples: **MQTT, CoAP, LwM2M** (lightweight machine to machine).




*M2M: machines talk directly or through a gateway (picture added).*

# B4. WSN, IEEE 802.15.4, SCADA and RFID

## B4.1 WSN protocols

- **Wireless Sensor Network** — sensors collect and transmit data.
- Examples: **Zigbee, 6LoWPAN, RPL**.

## B4.2 IEEE 802.15.4

- The basis of **Zigbee**: the **basic protocol for low-rate wireless personal area networks**.
- Works on **2.4 GHz / 868 MHz / 915 MHz**.
- **Short range and reliable** transmission.

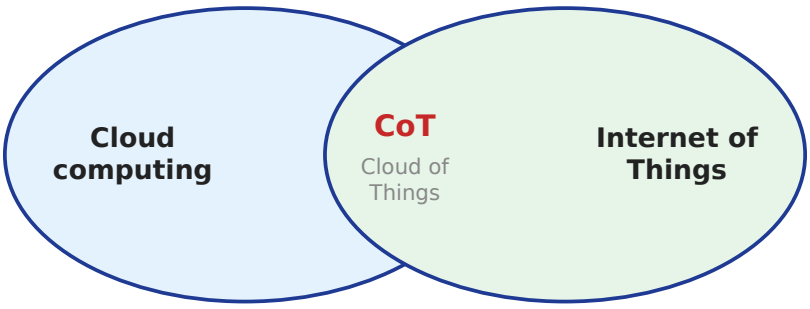
## B4.3 SCADA and RFID

| | SCADA | RFID |
|---|---|---|
| Full form | Supervisory Control and Data Acquisition | Radio Frequency Identification |
| Used for | Industrial control systems | Wireless identification |
| Where | Electricity grids and water systems | Communication between tags (like **FASTag**) and readers |
| Protocol | **DNP** (Distributed Network Protocol) is used in SCADA | — |

# B5. Cloud, Grid, SoA, MCC, IIoT and ARMET

## B5.1 CoT — Cloud of Things

- **Cloud computing + Internet of Things**.
- Data from IoT devices is **accessed, stored, processed and managed through platforms**.




*CoT = cloud computing + IoT (picture added).*

## B5.2 Grid computing

- A **distributed computing model** in which different resources (servers and storage) are **connected via a network** to perform **large-scale computation**.
- Resources are **shared** in this model.

## B5.3 SoA — Service Oriented Architecture

- An **architectural pattern that defines services**.
- Different services are **independent** and communicate with each other **via messages**.
- In SoA, **cloud platforms** are used to increase scalability and flexibility.

## B5.4 Cloud computing and cloud standards

- A model in which computing resources … (the rest of the sentence is cut off at the page edge of the notes; it describes computing resources provided as a service over the Internet).
- **Cloud standards** are important because they ensure **interoperability, security and portability**.

| Standard | Full form |
|---|---|
| **OCCI** | Open Cloud Computing Interface |
| **CSA** | Cloud Security Alliance |
| **ISO** | International Standard Organisation (International Organization for Standardization) |

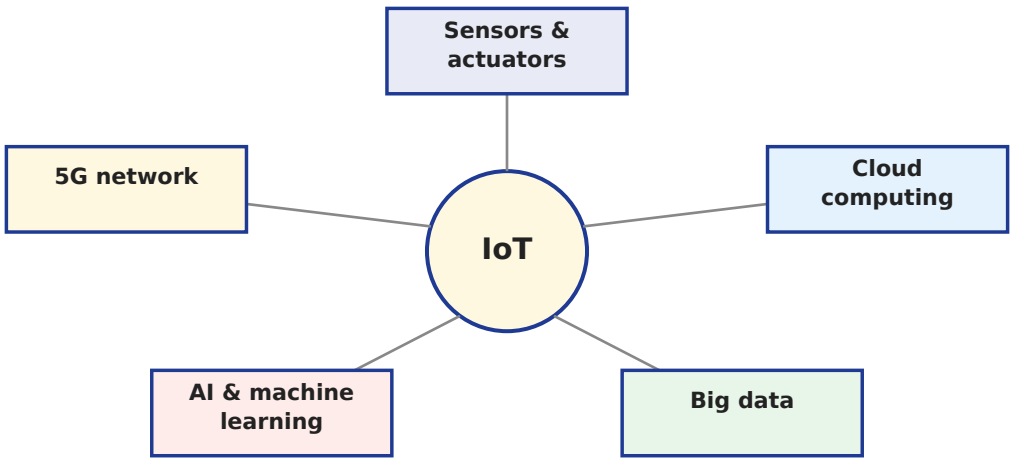
## B5.5 MCC — Mobile Cloud Computing

- **Connecting mobiles through cloud platforms**.
- Examples: **iCloud, Google Drive, Dropbox**.
- Folder sync.

## B5.6 IIoT — Industrial Internet of Things

- In industrial environments, **sensors, devices and machines are connected to the Internet for real-time monitoring**.
- These devices **collect and share data in real time** so industries can **optimize their operations**.
- **IIoT 4.0** is the **fourth industrial revolution**, in which physical machines are integrated with digital technologies.
- Its main focus is on **automation, data exchange and integrating other technologies**.

## B5.7 Basic technologies used in IoT




*The five basic technologies behind IoT.*

| # | Technology | Role |
|---|---|---|
| 1 | Sensors & actuators | Collect real-time data |
| 2 | Cloud computing | — |
| 3 | Big data | — |
| 4 | AI & machine learning | — |
| 5 | 5G network | — |

## B5.8 ARMET

- **Autonomous Risk Management & Engineering Technology**.

- Used to make **IoT systems secure and risk-free**.
- This approach **integrates safety and security**, in which **AI and machine learning** are used.

## B5.9 Role of IoT in a collaborative production environment

| # | Role |
|---|---|
| (i) | Increased autonomy and agility |
| (ii) | Resource management |
| (iii) | Synchronization |
| (iv) | Software agents |

# Appendix — Clarifications and reading notes

The handwritten pages were typed as faithfully as possible. These places were clarified, flagged or marked as added:

| Topic | In the handwritten notes | Here |
|---|---|---|
| AI definition | "… do not require human intelligence" | Kept; meaning: done without a human performing the task |
| First AI program | Logic Theorist, year hard to read, by Allen Newell | Kept (about 1956/57); the program was by Newell, Simon and Shaw |
| ELIZA | Credited to Joseph Engelberger | Flagged — ELIZA was written by Joseph Weizenbaum |
| Sophia | "UAE citizenship" in 2017 | Flagged — Saudi Arabia granted it |
| Deep Blue | "First intelligent humanoid" | Flagged — a chess computer, not a humanoid |
| Branches of AI | "Neutral logic" | Read as neural networks |
| Cloud computing definition | Sentence cut off at the page edge | Marked as incomplete |
| Role of IoT heading | Hard to read | Read as "Role of IoT in a collaborative production environment" |
| Added for revision | — | Capability tree, timeline, agent flows, reinforcement-learning loop, blockchain, fuzzy-logic, IoT layer, WoT, CoT, technology and M2M pictures; summary tables |